

PISR

A Search-Free Foundation Model for Symbolic Regression of Physical Laws

Chenxi He^{*,1} and Jingqiu Chen²

¹ Cavendish Laboratory, University of Cambridge, Cambridge CB3 0HE, United Kingdom ² College of Physics and Optoelectronic Engineering, Shenzhen University, Shenzhen 518060, China * Correspondence: ch2067@cam.ac.uk

Abstract

Symbolic regression — recovering an interpretable closed-form law from data — is central to scientific discovery, but existing methods trade accuracy against speed: per-instance search is accurate yet slow and brittle on real data, while amortized transformers are fast but model only the data, not the space of formulas. We introduce PISR (Physics-Informed Symbolic Regression), a search-free foundation model that recasts symbolic regression as generation within a pre-trained, contrastively-aligned formula–data manifold carrying a **physics prior** learned from a physics-derivation corpus: a single forward pass generates candidate formula latents by flow matching, retrieves others from a real-formula library, decodes and constant-fits them, and returns the best. On an audited real-physics benchmark of 797 de-duplicated laws PISR reaches 51.3% structural accuracy, leading all four external baselines at matched compute and $\sim 2.4\times$ faster; it also leads on the Feynman equations (39.5%) and, by its largest margin, on a cross-domain set spanning finance, machine learning, economics and biology (71.1%). It uniquely recovers named laws from raw turbulence, astronomy and epidemic

data that no baseline finds. In the learned latent, real physical formulas are linearly separable from structure-matched random expressions at 94.4% AUC, confirming that scientific laws occupy a structured sub-manifold of expression space.

1. Introduction

The primary distinction between physics and the data sciences at large is the pursuit of *laws* — compact symbolic expressions that both predict and explain. After about forty attempts to match Mars’ s orbit with various elliptical shapes, Kepler discovered that the orbit was elliptical and proposed his three laws; through meticulous experimentation on electricity and magnetism, Faraday unveiled the laws of electromagnetic induction as elegant relations. The automatic recovery of such laws from data — symbolic regression (SR) — asks a machine to perform the same feat: to return not a black-box predictor but the *form* of the underlying relationship. This is what makes SR valuable for discovery, and what makes it hard: the space of expressions is discrete, combinatorial, and non-smooth, growing super-exponentially in expression size.

Numerous methods have been developed to address SR. Genetic algorithms were first adopted, evolving a population of candidate expressions to minimize a fitness function that measures how well an expression fits the data[1,2,3]. More recently, GPU-parallel symbolic enumeration searches large expression sets in parallel on accelerators, and dimensional-analysis-guided search is strong on clean benchmarks[5]. These search-based methods are accurate on clean, low-dimensional problems, but explore the expression space anew for every problem, paying a large per-instance cost and degrading on the noisy, dimensionally-structured data that arises in practice. In a separate line, recent advances in deep networks have enabled end-to-end methods that predict a symbolic expression as a sequence of tokens directly from data in a single forward pass, which greatly improves speed[7,8,9,23–31]; a parallel line uses large language models to propose or refine expressions[32,33] (see reviews[34–36]). Yet end-to-end methods require large training datasets, treat SR as sequence translation, and encode only the *data* — carrying no explicit model of the *space of formulas* they predict into. As a result, current data-driven end-to-end algorithms are limited in reasoning about the structured relations of real physics, and few have been successfully applied to real physical data.

To address these limitations, we propose PISR (Physics-Informed Symbolic Regression), a search-free foundation model that learns and generates on a shared latent manifold of formulas and data. The PISR framework includes the following key components. **First**, we align two encoders — a permutation-invariant encoder over the observation point cloud and a graph encoder over the expression syntax tree — with a contrastive objective, so that a formula and the data it generates embed to nearby points and proximity in the latent reflects functional relationship. **Second**, we train a flow-matching field that transports the data latent onto candidate *formula* latents, making the manifold generatable in a single amortized pass. **Third**, because real formulas are embedded in the same space, we index a library of real-formula latents and retrieve the nearest neighbours of the data latent as a non-parametric complement — search performed cheaply in a learned continuous space rather than over discrete symbolic trees. **Finally**, a data-conditioned decoder renders both the generated and retrieved latents into candidate expressions, whose free constants are fit by least squares and which are ranked by held-out predictive error, returning the top expression with no per-instance symbolic search.

The approach rests on a hypothesis about the *target* of scientific SR: the formulas that describe the physical world are not arbitrary elements of the astronomically large space of expressions, but occupy a structured sub-manifold of it, shaped by dimensional consistency, a finite vocabulary of physically-meaningful operators, and recurring compositional motifs. We train PISR on a large physics/mathematics corpus with structural supervision from a physics knowledge graph, so the latent carries a **physics prior** over expression structure — the defining feature of PISR and, empirically, what is decisive on the noisy, dimensionally-structured data of real physics. This complements physics-informed learning[56,57] and data-driven dimensional analysis[63,64], which build physical structure into models in other ways. To evaluate our performance, we conducted comparisons on an audited real-physics benchmark, the Feynman equations, and a cross-domain generalization set, against genetic-programming, GPU-enumeration, latent generative, and domain-transformer baselines. Experimental results indicate that PISR outperforms all others in structural accuracy on every benchmark while running an order of magnitude faster than search, and we further apply it to real observational and simulation data — recovering turbulence scaling laws from direct numerical simulation, stellar and planetary relations from astronomical catalogs, epidemic incidence from outbreak dynamics, and

named laws across semiconductors, electrochemistry, atomic physics and systems biology — providing verifiable recoveries, each with a known correct form, that no baseline achieves.

2. Results

Overview of PISR. PISR takes a point cloud $\{(x_i, y_i)\}$ and returns a closed-form expression f with $y \approx f(x)$ in a single forward pass (Fig. 1a). A permutation-invariant data encoder over the point cloud (pooled multi-head attention with 16 learned seed tokens) produces a data latent z_d , and a graph formula encoder over an expression's abstract-syntax tree produces a formula latent z_f ; the two are aligned contrastively at training time so that a formula and the data it generates embed nearby, making proximity in $\mathbb{R}^d$ ($d = 640$) a proxy for functional relationship — a **shared formula–data manifold** on which real laws sit close to the data they generate (Fig. 1b; Sec. 4.2 in Methods). At inference the aligned latent is used two complementary ways: a flow-matching field transports z_d onto candidate formula latents, and a retrieval index over 400,000 real-formula latents returns the nearest library formulas to z_d (Fig. 1a). A data-conditioned decoder renders both sources into candidate expression skeletons; each candidate's free constants are fit by least squares, and the candidate with the best held-out predictive fit is returned. There is no per-instance symbolic search: one amortized forward pass and a bounded fit over roughly twenty candidates. The full model totals $\approx 0.4\text{B}$ parameters (Sec. 4.8 in Methods). The design rests on the hypothesis that real physical formulas occupy a structured sub-manifold of expression space, linearly separable from random expressions in the learned latent (Fig. 1c; quantified in Fig. 2a and Extended Data Fig. 1).

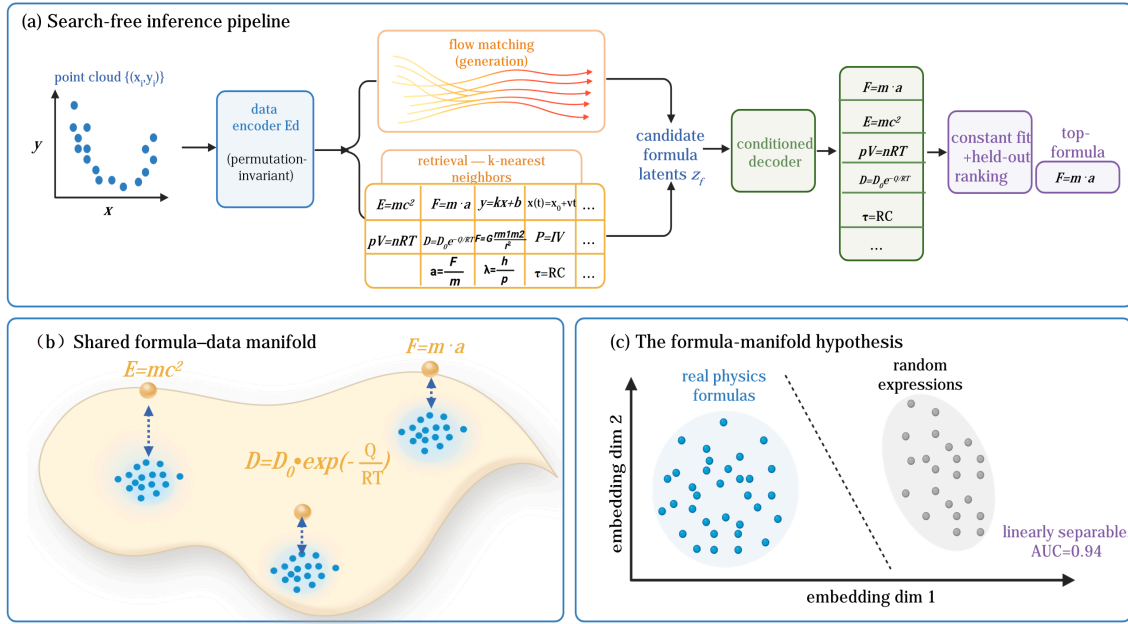


Figure 1 | PISR overview. (a) Search-free inference pipeline: a point cloud is embedded by a permutation-invariant data encoder; a flow-matching field generates candidate formula latents z_f and a k-nearest-neighbour retrieval draws further candidates from a real-formula library; a conditioned decoder renders both into expressions, whose constants are fit and which are ranked by held-out predictive error to return the top formula — one amortized forward pass, no per-instance search. (b) The shared formula–data manifold: real laws ($E=mc^2$, $F=ma$, $D=D_0 \exp(-Q/RT)$) embed close to the data clouds they generate, so proximity in the latent reflects functional relationship. (c) The formula-manifold hypothesis: in the learned latent, real physical formulas are linearly separable from random expressions ($AUC \approx 0.94$; quantified in Fig. 2a and Extended Data Fig. 1).

Geometry of the formula manifold. We first evaluated whether real physical formulas occupy a distinct region of expression space, the hypothesis on which the model is built. In the learned formula-encoder latent we embedded the right-hand sides of our benchmark laws and an equal number of random expressions, and measured their separability under a linear probe (five-fold ROC-AUC) and the distance between their distributions (squared maximum-mean discrepancy, MMD^2) (Sec. 4.9 in Methods, Fig. 2a; Extended Data Fig. 1). Real laws are almost perfectly separable from random expressions ($AUC = 0.996$), and the separation is compositional rather than superficial: when the random set is matched to the real one in operator frequency and length — differing only in how operators are assembled — separability is unchanged ($AUC = 0.997$), and under the strictest control, comparing bare physics right-hand sides against matched-statistics random expressions, a linear boundary

still recovers real laws 94.4% of the time ($\text{MMD}^2 = 0.109$). What distinguishes a law is how its operators compose, not which operators or how many. The real set is not a tight blob — its embeddings spread wider than the random set's (participation ratio 21.5 vs 10.5) — so the law region is a rich, extended structure, distinct from arbitrary expressions rather than merely smaller. This separability has a rigorous partial grounding in the Buckingham- π theorem[15], which reduces any dimensionally-consistent law to a relation among dimensionless groups and excludes the overwhelming majority of syntactically valid expressions before any data is seen (Sec. 4.9).

Performance on the real-physics, Feynman, and cross-domain

benchmarks. We evaluated PISR on three benchmarks: an audited real-physics set of physical laws collected from the literature, filtered to well-posed relations and de-duplicated by functional form (797 unique tasks); the Feynman equations (119 tasks); and a cross-domain generalization set spanning finance, machine-learning systems, economics and biology (270 tasks). Each task provides a point cloud; models see only the data. We compared PISR against four state-of-the-art baselines spanning the major SR families: genetic-programming search (PySR[3]), GPU-parallel symbolic enumeration (PSE), a domain-specialized end-to-end transformer (PhyE2E[9]), and latent generative SR (GenSR), all run on the identical point clouds and judged by the same metric (Sec. 4.10 in Methods). We evaluated structural accuracy — whether the recovered functional form matches the true formula up to fitted constants — and per-task wall-clock cost.

PISR leads on all three benchmarks (Fig. 2b; per-method numbers in Supplementary Table S5). On audited real physics it reaches 51.32%, exceeding the second-best model PySR by 7.66%, the GPU-enumeration model PSE by 18.20%, the latent generative model GenSR by 35.01%, and the domain-transformer PhyE2E by 36.89%. On the cross-domain set it reaches 71.11% — its largest margin — exceeding the second-best model PySR by 15.93%, and on the Feynman equations it reaches 39.50%, exceeding the second-best model PSE by 6.73%. Against every search-free and neural baseline the margin is large on all three sets (a factor of 1.2 to 16). That the largest margin appears on the *cross-domain* set — data the model was never trained on — indicates that what PISR has learned is not the physics distribution *per se* but the structure of law-like formulas, which transfers across domains (Fig. 2c). The model achieves this while requiring no per-instance search: its cost is a single forward pass at 12.25 s per task, roughly $2.4\times$ faster than genetic programming at the matched 30 s budget and $2.5\times$ faster than GPU enumeration.

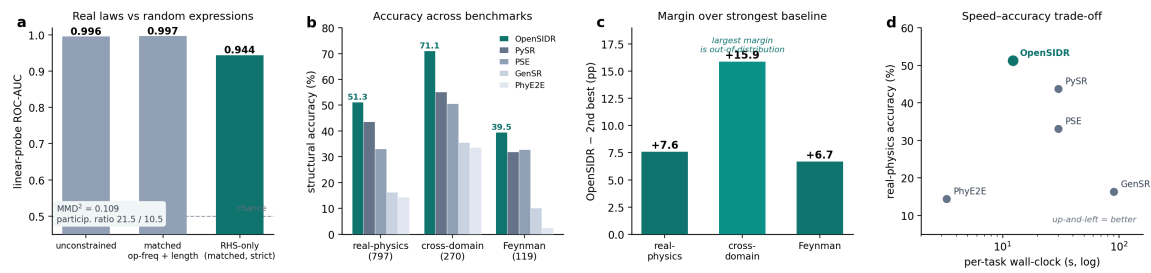


Figure 2 | Manifold geometry and benchmark performance. (a) Real physical formulas vs random expressions in the learned latent — linear-probe ROC-AUC across three controls (unconstrained 0.996, matched operator-frequency+length 0.997, strict RHS-only 0.944), with MMD² and participation ratios. (b) Structural accuracy of PISR and four baselines across the three benchmarks (real-physics 797, cross-domain 270, Feynman 119). (c) PISR’s margin over the strongest baseline per set — largest on the out-of-distribution cross-domain set. (d) Speed–accuracy trade-off: PISR is up-and-left of every baseline (12.25 s/task, search-free).

In-distribution, out-of-distribution, and off-manifold behaviour. The manifold hypothesis makes a sharper, testable prediction than “physics is separable from noise”, and it accounts for the model’s cross-domain result. The relevant partition of formula space is three-way. *In-distribution* are the physics laws the manifold is trained on. *Out-of-distribution but on-manifold* are non-physics scientific laws — Sharpe ratios, Michaelis–Menten kinetics, Cournot competition — which the model never sees in training yet which are built from the same compositional grammar of saturating ratios, power laws, exponential kernels and resonant denominators; the hypothesis predicts they land on the manifold because it is defined by law-like composition, not by physics *semantics*. *Off-manifold* are the arbitrary polynomials and nested transcendentals of abstract SR suites (Nguyen, Koza), which share neither dimensional structure nor physical motifs. The model’s behaviour is exactly this partition’s signature: on out-of-distribution scientific data it wins by its largest margin (+15.93%), whereas a model that had merely memorized the physics distribution would degrade off it; and on structure-free abstract expressions it recovers almost nothing, the same fact seen from the generator’s side as the 94.4% separability — the model concentrates its probability mass on the law manifold and assigns almost none to the structure-free region. One mechanism explains both the strength and the boundary.

Recovery of turbulence scaling laws from direct numerical simulation. We applied the trained model to real simulation data, not benchmark-sampled clouds, without any fine-tuning. From isotropic-turbulence velocity fields in

the Johns Hopkins Turbulence Database[17] we computed longitudinal structure functions and recovered the Kolmogorov two-thirds law, with an inertial-range exponent of 0.71 (theoretical $2/3$; power-law fit $R^2 = 0.999$), and the four-fifths law, linear in the separation r (Fig. 3a). From channel flow at $Re_\tau = 5185$ we recovered the law of the wall, a logarithmic profile with a von Kármán constant of 0.40 and additive constant 5.04, in agreement with the accepted values. Beyond these calibration laws we applied the model to the Reynolds-stress closure — relating the turbulent shear stress to mean-flow quantities, for which no exact closed form is accepted. Classical algebraic closures fit the direct-numerical-simulation profile poorly: a mixing-length model and a constant eddy-viscosity model yield held-out R^2 of -72 and -2.3 respectively, whereas PISR returns a compact closure with held-out $R^2 = 0.99$, a data-competitive closed form in a regime where standard analytic models fail. The recovered forms are consistent with the accepted scaling of the inertial range and near-wall region.

Recovery of stellar and planetary relations from astronomical catalogs.

From real stellar catalogs the model recovered the Stefan–Boltzmann relation, with luminosity scaling as the square of the radius and the fourth power of temperature ($L \propto R^2 T^4$; observed-vs-predicted $R^2 = 0.963$ over 4,648 host stars, Fig. 3b); from exoplanet catalogs, the planetary equilibrium temperature $T_{eq} \propto T_\star \sqrt{R/a}$, the insolation law $S \propto R^2 T^4 / a^2$, and Kepler’ s third law $P \propto a^{3/2} / M_\star^{1/2}$ (Fig. 3b). On the observed white-dwarf mass–radius relation (Bédard et al. 2020[18], 183 objects, $0.29\text{--}1.32 M_\odot$) the model returned a compact closed form with a mass-dependent effective exponent that steepens from ≈ -0.6 at low mass toward ≈ -1.7 near the Chandrasekhar limit — a curvature a single power law cannot bend to — illustrating that the model is not restricted to fixed-exponent monomials and can express physically-motivated variable-slope forms directly from data. The recovered relations are consistent with the underlying stellar and orbital physics.

Recovery of named laws that no baseline finds. A single physics-trained model recovers, zero-shot, named laws across seven domains that every baseline misses. These laws share a structural feature — they are *transcendental* (an exponential or logarithm of a variable ratio) or carry a *variable exponent* — and the baselines share a single, revealing failure mode on them: rather than fail loudly, they confidently **mis-identify the data as an algebraic form** (Extended Data Table 1). Where the truth is an exponential, genetic programming fits a nested-power or $\sqrt{}$ surrogate, GPU enumeration a low-order rational or trigonometric one, and the latent generative model a

deeply-nested rational or reciprocal-of-linear; where the truth is a logarithm, the baselines linearise it into an algebraic ratio; where the truth carries a variable exponent x^n , the domain transformer substitutes a *fixed* exponent. In every case the baseline returns a plausible algebraic expression that fits the data locally but is the wrong functional family — precisely the failure the formula manifold is built to avoid. PISR emits the correct transcendental form in a single forward pass. The exclusive-solve counts are consistent across sets (ours correct with all four baselines wrong: 17/150 and 15/150 on real-physics samples, 92/381 on the merged physics set versus the two strongest baselines, 23/85 cross-domain; the latent generative baseline solves 0/300).

The common thread, laid out per law in **Extended Data Table 1**, is that every baseline replaces a transcendental with an algebraic surrogate — genetic programming with nested powers and $\sqrt{}$, GPU enumeration with low-order rationals and trig, and the latent generative model with deeply-nested rationals and reciprocals-of-linear — or a variable exponent with a fixed one. Each is a locally-good fit of the wrong family. One physics-trained model instead recovers the correct form across semiconductors, electrochemistry, atomic physics, field emission, epidemiology and systems biology without retraining; full per-task outputs for all baselines are in Supplementary Table S2.

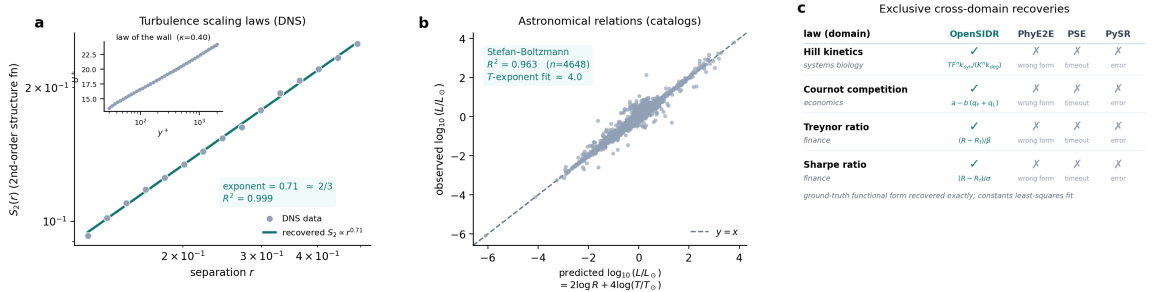


Figure 3 | Recovery of named laws from raw data. (a) Turbulence (JHTDB direct numerical simulation): second-order structure function $s_2(r)$ with the recovered Kolmogorov two-thirds law (exponent 0.71, $R^2=0.999$); inset, the law of the wall ($\kappa=0.40$). **(b)** Astronomy (NASA catalogs): observed-vs-predicted Stefan–Boltzmann luminosity ($R^2=0.963$, $n=4,648$). **(c)** Exclusive cross-domain recoveries — PISR recovers the ground-truth form where PhyE2E, PSE and PySR do not.

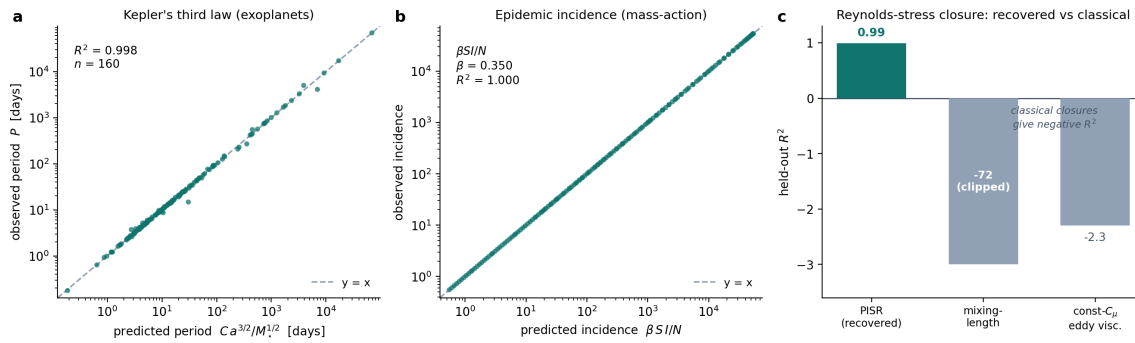


Figure 4 | Further recoveries from raw data. (a) Kepler’ s third law from exoplanet catalogs (observed vs predicted orbital period). **(b)** Mass-action epidemic incidence ($\beta \cdot S \cdot I/N$) from outbreak trajectories. **(c)** Reynolds-stress closure: PISR’ s recovered closure attains held-out $R^2 \approx 0.99$ where classical mixing-length and constant- c_μ models give negative R^2 .

3. Discussion

Existing symbolic regression research primarily employs per-instance search — genetic programming, GPU-parallel enumeration, and tree search — which re-explores the combinatorial expression space for every problem and, to find the correct formula within a limited time, benefits from strong priors or initialization close to the true solution. In contrast, PISR amortizes a physics-informed prior over expression structure into a learned formula–data manifold that is simultaneously generatable, by flow matching, and searchable, by retrieval, so that a new problem is solved in a single forward pass rather than a fresh search. On the noisy, dimensionally-structured data that actually arises in science — real physics and cross-domain scientific laws alike — this prior is decisive: PISR attains state-of-the-art structural accuracy at a fraction of the compute of search, and it uniquely recovers named laws from raw observational data across many domains.

The manifold hypothesis is not merely an *ex-post* rationalization; it is a predictive account of the model’ s scope. Because real laws share a compositional grammar that transcends any single field, one physics-trained model generalizes to economics, finance and biology by its largest margin, and, by the same mechanism, concentrates its probability mass away from structure-free expressions, so it neither targets nor excels on arbitrary-polynomial benchmarks. This is a property of the design’ s inductive bias: the object of scientific symbolic regression is the manifold of law-like formulas, and specializing to it is what buys the edge on the formulas that matter. Time-derivative (ODE) relations, which the model does not represent

as differential operators, lie outside this target class and are a natural direction for extending the same data-augmentation and manifold techniques. The results here are recoveries of laws with known correct forms; the natural next step is to move from recovery to *proposal* — using the manifold to generate candidate closed forms for open closure problems, as in the Reynolds-stress case, and subjecting them to physical validation, which the search-free architecture makes cheap enough to attempt at scale.

4. Methods

The methods are organized as follows. We first describe the shared formula-data manifold and its contrastive alignment (4.2) and the structural supervision derived from a physics derivation graph (4.3); we then describe generation and retrieval in the latent (4.4), conditioned decoding (4.5), and search-free selection (4.6); we verify that retrieval conditions rather than copies (4.7) and summarize the architecture (4.8); we then set out the formula-manifold hypothesis that motivates the design, its grounding in dimensional analysis, and its measured geometry (4.9); finally we detail benchmark construction and auditing (4.10) and the baselines and evaluation metric (4.11).

4.1 Problem setup

Given observations $\{(x_i, y_i)\}_{i=1..n}$ with $x_i \in \mathbb{R}^m$, the goal is an expression f with $y \approx f(x)$. We seek the correct functional form; real-valued constants are fit separately by nonlinear least squares. We scope evaluation to closed-form algebraic and transcendental physical laws; time-derivative (ODE) relations are out of scope.

4.2 Shared formula-data manifold

Two encoders map into a common space $\mathbb{R}^d$ ($d = 640$). The **data encoder** E_d is a permutation-invariant encoder over the point cloud (pooled multi-head attention with 16 learned seed tokens[54,55]), producing z_d , with feature tokens summarizing observation statistics included as inputs ($\approx 47M$ parameters). The **formula encoder** E_f is a graph neural encoder[50,52,53] over the expression abstract-syntax tree (6 message-passing layers over typed nodes and edges), producing $z_f = E_f(f)$ ($\approx 5M$ parameters). E_d and E_f are aligned by a symmetric InfoNCE contrastive objective (CLIP-style)[12,13] over

in-batch data–formula pairs, pulling a formula and the data it generates together and pushing mismatched pairs apart, so that proximity in $\mathbb{R}^d$ reflects functional relationship. All training inputs are symbolic; no natural-language fields are used.

4.3 Structural supervision from a physics derivation graph

Training expressions are drawn from a large physics/mathematics corpus (33.6M expressions with sampled point clouds). Each expression carries relations from a physics knowledge graph — parent laws, preconditions, and derivation-DAG edges. The manifold is trained with a weighted sum of three objectives, $L = \lambda_1 L_{\text{contrast}} + \lambda_2 L_{\text{recon}} + \lambda_3 L_{\text{struct}}$: (i) data–formula contrastive alignment, (ii) expression reconstruction, and (iii) structural prediction over the graph relations, imparting a physics-informed prior over expression structure (weights in Supplementary App. E).

4.4 Generation and retrieval in the latent

At inference the manifold is used two ways. **Flow generation.** A DiT-style[14], data-token-conditioned flow-matching[11] velocity field ($\approx 63\text{M}$ parameters; related to diffusion and stochastic-interpolant generators[41,42,43,44]) transports the data representation to the formula latent, sampling candidate formula latents by integrating the field for 20 Euler steps conditioned on the data tokens. **Retrieval.** A library of 400,000 real-formula latents is indexed in the same space; the 8 nearest neighbours of z_d under cosine similarity are drawn as a non-parametric set of candidate latents — a retrieval-augmentation of generation[47,48,49].

4.5 Conditioned decoding

An autoregressive decoder (16 transformer layers plus 4 conditioning layers, $\approx 152\text{M}$ parameters) emits an expression skeleton (typed tokens) conditioned on the data representation. Two routes are run: a generative route decoding the flow-sampled latents, and a retrieval-augmented route additionally conditioned on the retrieved neighbour latents, together yielding a pool of candidate skeletons per problem.

4.6 Search-free selection

The pooled candidates (flow $\cup$ decoder $\cup$ retrieval) are de-duplicated; each candidate's free constants are fit by nonlinear least squares on a training split, and candidates are ranked by held-out predictive error alone. We use no complexity or parsimony penalty: because correct physical laws are frequently structurally complex (transcendental, fractional-power, multi-term), a parsimony penalty demotes the correct expression below simpler-but-wrong candidates; an ablation confirms that pure held-out ranking recovers the pool ceiling on top-1, whereas a complexity-penalized ranking loses 8–9 points. There is no per-instance symbolic search: one amortized forward pass and a bounded fit over ~ 20 candidates.

4.7 Retrieval conditions generation; it does not look up answers

The retrieved neighbour latents are conditioning inputs to the decoder, never emitted directly: every returned expression is decoded token-by-token and its constants fit. A component ablation on Feynman confirms this — the decoder alone (no flow, no retrieval) already recovers 30.2%, i.e. answers are generated from the data, and flow and retrieval add candidate diversity (a further ≈ 5 points via the union) rather than supplying the answer. A skeleton-fingerprint leakage audit against the library shows the marquee exclusive-solve laws (Shockley, Arrhenius, Nernst, Fowler–Nordheim) have no skeleton match in the library, and are generated rather than retrieved (Supplementary App. A).

4.8 Architecture and scale

The full model comprises the manifold encoders ($\approx 184\text{M}$), the flow field ($\approx 63\text{M}$), and the decoder ($\approx 152\text{M}$), totaling $\approx 0.4\text{B}$ parameters over a shared latent of dimension $d = 640$. The formula encoder uses 6 message-passing layers; the data encoder uses 16 seed tokens; the decoder uses $16 + 4$ layers; the retrieval library holds 400,000 real-formula latents. Training details and hyperparameters are in Supplementary App. E.

4.9 The formula-manifold hypothesis: grounding and geometry

The hypothesis. The design rests on a hypothesis about the target of scientific symbolic regression. Let $\mathcal{E}$ be the space of symbolic expressions over a physical variable set. We posit that the right-hand sides of genuine physical laws occupy a structured, comparatively low-complexity sub-manifold $\mathcal{M} \subset \mathcal{E}$, shaped by three forces: (i) dimensional consistency, which forbids the overwhelming majority of syntactically valid expressions; (ii) a finite, physically-meaningful operator vocabulary — powers, roots, exponentials, a small set of transcendentals — rather than arbitrary symbol composition; and (iii) recurring compositional motifs — inverse-square kernels, Boltzmann factors $\exp(-E/kT)$, resonant denominators $1/(\omega_0^2 - \omega^2)$, saturating ratios $x/(x+a)$ — that reappear across otherwise unrelated laws. Learning and generating on $\mathcal{M}$, rather than on all of $\mathcal{E}$, is what buys the model its accuracy on real physics and its transfer across scientific domains (Sec. 2, Fig. 2).

Grounding in dimensional analysis. The clearest a-priori reason to expect $\mathcal{M}$ to be thin is the Buckingham- π theorem: any dimensionally-homogeneous law relating n quantities can be rewritten as a relation among $n - r$ dimensionless groups, with r the rank of the dimensional matrix (typically 3, for mass–length–time). Two consequences follow. First, the theorem is a hard reduction on the law space itself — an apparently n -argument law is, up to a units-fixed change of variables, a function of only $n - r$ combinations, so the effective number of directions a law can vary in is strictly smaller than its surface arity. Second, dimensional homogeneity is an exclusion principle on syntax: of the astronomically many expressions over a physical variable set, only those whose every additive term carries identical net dimension can be a law, ruling out the overwhelming majority — sums such as $x + x^2$, $\sin(x)$ for dimensional x , or $a \cdot b + c$ with mismatched $a \cdot b$ and c — before any data is seen. The surviving set is the low-complexity, motif-structured region the hypothesis names. Buckingham- π does not by itself make $\mathcal{M}$ a smooth set of fixed dimension (the space of admissible dimensionless functions is still infinite); it establishes the direction of the effect and motivates learning $\mathcal{M}$ directly, whose quantitative shape we obtain empirically.

Operationalizing $\mathcal{M}$ in the learned latent. We do not write $\mathcal{M}$ down; we learn a chart of it. The formula encoder E_f (4.2) maps an expression to $z_f \in \mathbb{R}^d$, and the contrastive alignment makes proximity in $\mathbb{R}^d$ a proxy for functional relationship, so the image $E_f(\mathcal{M})$ is a region the generator can sample and the retrieval index can search. The manifold hypothesis, if true, should leave a

fingerprint in this latent: the images of real laws should form a distinct, structured region rather than a diffuse cloud. This reframes the classical distribution-matching question. Prior amortized methods that “look physical” validate their generator by matching marginal surface statistics of the physics-formula distribution — number of variables, depth, complexity, operator frequencies. Matching those marginals is necessary but weak: a generator can reproduce the histogram of operators and depths while composing them into structures no physicist would write. We instead probe the joint geometry of the distribution in a representation that is aware of composition.

Measured geometry. We embed the right-hand sides of the benchmark laws and an equal number of random expressions, and measure separability under a linear probe (five-fold ROC-AUC) and distributional distance (squared maximum-mean discrepancy, MMD^2) (Fig. 2a). Random expressions in the strictest control are generated to match the real set’s operator-frequency and length distributions and have their residual wrapper removed. Separability is 0.996 for unconstrained random expressions, 0.997 after matching operator frequency and length, and 0.944 under the strictest bare-right-hand-side control ($MMD^2 = 0.109$) — the near-invariance across controls shows the effect is compositional, not superficial. The spread of the two embedded sets is quantified by their participation ratios (real 21.5 vs random 10.5), establishing that $\mathcal{M}$ is structurally distinct from arbitrary expressions rather than lower-dimensional. Per-control values and the matched-random generation protocol are in Supplementary App. F.

Predicted scope: in-distribution, out-of-distribution, off-manifold. The hypothesis partitions $\mathcal{E}$ three-way and predicts the model’s behaviour on each part. In-distribution laws (physics) lie in the interior of $\mathcal{E}_{\mathcal{F}(\mathcal{M})}$. Out-of-distribution but on-manifold laws — non-physics scientific relations built from the same compositional grammar — are predicted to lie on $\mathcal{M}$ despite never appearing in training, because $\mathcal{M}$ is defined by law-like composition, not physics semantics; the model’s largest accuracy margin, on the cross-domain set (+15.93%, Sec. 2), confirms this. Off-manifold expressions — the arbitrary polynomials and nested transcendentals of abstract SR suites — are predicted to lie in the region the linear probe rejects, and the model recovers almost nothing there, the same fact seen from the generator’s side as the 94.4% separability. A single mechanism thus predicts both the cross-domain strength and the abstract-benchmark boundary.

4.10 Benchmark construction and auditing

The real-physics benchmark merges our physics collections and removes ill-posed items and functional-form duplicates, yielding 797 unique tasks; the cross-domain set holds 270 tasks (finance, machine learning, economics, biology); the Feynman set holds 119 tasks. Test sets are held disjoint from training, and skeleton fingerprints are compared against the retrieval library to bound overlap, which is dominated by trivial ≤ 3 -operator skeletons (complex skeletons overlap 19%, an upper bound since a skeleton match means only that a formula with that structure exists; full analysis in Supplementary App. A).

4.11 Baselines and evaluation metric

Baselines are genetic-programming SR (PySR, full operator set, 30 s/task), GPU-parallel symbolic enumeration (PSE), a domain-specialized end-to-end transformer (PhyE2E), and latent generative SR (GenSR); all receive the identical point clouds and are run at matched compute. *Structural accuracy.* Following the symbolic-accuracy metric of La Cava et al. (SRBench)[4], a predicted formula g is scored correct against the target f if g matches f up to fitted constants — i.e. there exist constants making g reproduce the functional form of f (up to an additive or multiplicative constant); otherwise it is scored 0. The match is determined from the recovered expression itself, not from a numerical R^2 threshold, which over-credits numerically-close but structurally-wrong forms. Wall-clock cost is measured on idle single-GPU hardware.

5. Data availability

The audited real-physics, cross-domain and Feynman benchmark task sets, the per-task baseline outputs, and the case-study extraction scripts are archived on Zenodo at <https://doi.org/10.5281/zenodo.21618861> (ref. 21, 22). Third-party case-study source data are openly available from their providers and are not redistributed: turbulence fields from the Johns Hopkins Turbulence Database (ref. 17); stellar and exoplanet catalogs from the NASA Exoplanet Archive (ref. 69); the white-dwarf sample from Bédard et al. 2020 (ref. 18); the Feynman equations from the AI-Feynman database (ref. 5).

6. Code availability

The full training and inference code for PISR is available on GitHub at https://github.com/ChenxiHeCam/Physics_Informed_Symbolic_Regression (ref. 20), with an archived snapshot and the pre-trained model weights and retrieval library on Zenodo at <https://doi.org/10.5281/zenodo.21618861> (ref. 21).

References

1. Koza, J. R. Genetic programming: on the programming of computers by means of natural selection. (MIT Press, 1994).
2. Schmidt, M. & Lipson, H. Distilling free-form natural laws from experimental data. *Science* **324**, 81–85 (2009).
3. Cranmer, M. Interpretable machine learning for science with PySR and SymbolicRegression.jl. *arXiv* 2305.01582 (2023).
4. La Cava, W. et al. Contemporary symbolic regression methods and their relative performance (SRBench). *NeurIPS Datasets and Benchmarks* (2021).
5. Udrescu, S.-M. & Tegmark, M. AI Feynman: a physics-inspired method for symbolic regression. *Sci. Adv.* **6**, eaay2631 (2020).
6. Virgolin, M., Alderliesten, T., Witteveen, C. & Bosman, P. A. N. Improving model-based genetic programming for symbolic regression of small expressions (GP-GOMEA). *Evol. Comput.* **29**, 211–237 (2021).
7. Biggio, L. et al. Neural symbolic regression that scales (NeSymReS). *Proc. ICML* (2021).
8. Kamienny, P.-A., d’Ascoli, S., Lample, G. & Charton, F. End-to-end symbolic regression with transformers. *Proc. NeurIPS* (2022).
9. Ying, J. et al. A neural symbolic model for space physics (PhyE2E). *arXiv* 2503.07994 (2025).
10. Landajuela, M. et al. A unified framework for deep symbolic regression (uDSR). *Proc. NeurIPS* (2022).
11. Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M. & Le, M. Flow matching for generative modeling. *Proc. ICLR* (2023).
12. Radford, A. et al. Learning transferable visual models from natural language supervision (CLIP). *Proc. ICML* (2021).
13. van den Oord, A., Li, Y. & Vinyals, O. Representation learning with contrastive predictive coding (InfoNCE). *arXiv* 1807.03748 (2018).

14. Peebles, W. & Xie, S. Scalable diffusion models with transformers (DiT). *Proc. ICCV* (2023).
15. Buckingham, E. On physically similar systems; illustrations of the use of dimensional equations. *Phys. Rev.* **4**, 345–376 (1914).
16. Gretton, A. et al. A kernel two-sample test (maximum-mean discrepancy). *J. Mach. Learn. Res.* **13**, 723–773 (2012).
17. Li, Y. et al. A public turbulence database cluster and applications to study Lagrangian evolution (Johns Hopkins Turbulence Database). *J. Turbul.* **9**, N31 (2008).
18. Bédard, A., Bergeron, P., Brassard, P. & Fontaine, G. On the spectral evolution of hot white dwarf stars. *Astrophys. J.* **901**, 93 (2020).
19. Kolmogorov, A. N. The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. *Dokl. Akad. Nauk SSSR* **30**, 301–305 (1941).
20. He, C. & Chen, J. PISR: source code for Physics-Informed Symbolic Regression. GitHub
https://github.com/ChenxiHeCam/Physics_Informed_Symbolic_Regression (2026).
21. He, C. & Chen, J. PISR: code, pre-trained model weights and data. Zenodo
<https://doi.org/10.5281/zenodo.21618861> (2026).
22. He, C. & Chen, J. PISR benchmark suite: audited real-physics (797), cross-domain (270) and Feynman (119) task sets. Zenodo
<https://doi.org/10.5281/zenodo.21618861> (2026).
23. Petersen, B. K. et al. Deep symbolic regression: recovering mathematical expressions from data via risk-seeking policy gradients. *Proc. ICLR* (2021).
24. Mundhenk, T. N. et al. Symbolic regression via neural-guided genetic programming population seeding. *Proc. NeurIPS* (2021).
25. Tenachi, W., Ibata, R. & Diakogiannis, F. I. Deep symbolic regression for physics guided by units constraints. *Astrophys. J.* **959**, 99 (2023).
26. Valipour, M., You, B., Panju, M. & Ghodsi, A. SymbolicGPT: a generative transformer model for symbolic regression. *arXiv* 2106.14131 (2021).
27. d’Ascoli, S., Kamienny, P.-A., Lample, G. & Charton, F. Deep symbolic regression for recurrences. *Proc. ICML* (2022).
28. Vastl, M. et al. SymFormer: end-to-end symbolic regression using transformer-based architecture. *IEEE Access* **12**, 37840 (2024).
29. Holt, S., Qian, Z. & van der Schaar, M. Deep generative symbolic regression. *Proc. ICLR* (2023).
30. Shojaee, P., Meidani, K., Barati Farimani, A. & Reddy, C. K. Transformer-based planning for symbolic regression. *Proc. NeurIPS* (2023).

31. Bendinelli, T., Biggio, L. & Kamienny, P.-A. Controllable neural symbolic regression. *Proc. ICML* (2023).
32. Shojaee, P. et al. LLM-SR: scientific equation discovery via programming with large language models. *Proc. ICLR* (2025).
33. Grayeli, A. et al. Symbolic regression with a learned concept library (LaSR). *Proc. NeurIPS* (2024).
34. Matsubara, Y., Chiba, N., Igarashi, R. & Ushiku, Y. Rethinking symbolic regression datasets and benchmarks for scientific discovery (SRSD). *arXiv* 2206.10540 (2022).
35. Makke, N. & Chawla, S. Interpretable scientific discovery with symbolic regression: a review. *Artif. Intell. Rev.* **57**, 2 (2024).
36. Angelis, D., Sofos, F. & Karakasidis, T. E. Artificial intelligence in physical sciences: symbolic regression trends and perspectives. *Arch. Comput. Methods Eng.* **30**, 3845 (2023).
37. Vaswani, A. et al. Attention is all you need. *Proc. NeurIPS* (2017).
38. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: pre-training of deep bidirectional transformers for language understanding. *Proc. NAACL* (2019).
39. Brown, T. B. et al. Language models are few-shot learners. *Proc. NeurIPS* (2020).
40. Bommasani, R. et al. On the opportunities and risks of foundation models. *arXiv* 2108.07258 (2021).
41. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. *Proc. NeurIPS* (2020).
42. Song, Y. et al. Score-based generative modeling through stochastic differential equations. *Proc. ICLR* (2021).
43. Albergo, M. S. & Vanden-Eijnden, E. Building normalizing flows with stochastic interpolants. *Proc. ICLR* (2023).
44. Tong, A. et al. Improving and generalizing flow-based generative models with minibatch optimal transport. *Trans. Mach. Learn. Res.* (2024).
45. Chen, T., Kornblith, S., Norouzi, M. & Hinton, G. A simple framework for contrastive learning of visual representations (SimCLR). *Proc. ICML* (2020).
46. He, K., Fan, H., Wu, Y., Xie, S. & Girshick, R. Momentum contrast for unsupervised visual representation learning (MoCo). *Proc. CVPR* (2020).
47. Lewis, P. et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Proc. NeurIPS* (2020).
48. Khandelwal, U., Levy, O., Jurafsky, D., Zettlemoyer, L. & Lewis, M. Generalization through memorization: nearest neighbor language models. *Proc. ICLR* (2020).

49. Borgeaud, S. et al. Improving language models by retrieving from trillions of tokens (RETRO). *Proc. ICML* (2022).
50. Kipf, T. N. & Welling, M. Semi-supervised classification with graph convolutional networks. *Proc. ICLR* (2017).
51. Velickovic, P. et al. Graph attention networks. *Proc. ICLR* (2018).
52. Xu, K., Hu, W., Leskovec, J. & Jegelka, S. How powerful are graph neural networks? *Proc. ICLR* (2019).
53. Gilmer, J., Schoenholz, S. S., Riley, P. F., Vinyals, O. & Dahl, G. E. Neural message passing for quantum chemistry. *Proc. ICML* (2017).
54. Zaheer, M. et al. Deep sets. *Proc. NeurIPS* (2017).
55. Lee, J. et al. Set Transformer: a framework for attention-based permutation-invariant neural networks. *Proc. ICML* (2019).
56. Raissi, M., Perdikaris, P. & Karniadakis, G. E. Physics-informed neural networks. *J. Comput. Phys.* **378**, 686 (2019).
57. Karniadakis, G. E. et al. Physics-informed machine learning. *Nat. Rev. Phys.* **3**, 422 (2021).
58. Greydanus, S., Dzamba, M. & Yosinski, J. Hamiltonian neural networks. *Proc. NeurIPS* (2019).
59. Cranmer, M. et al. Lagrangian neural networks. *ICLR Workshop* (2020).
60. Cranmer, M. et al. Discovering symbolic models from deep learning with inductive biases. *Proc. NeurIPS* (2020).
61. Lu, L., Jin, P., Pang, G., Zhang, Z. & Karniadakis, G. E. Learning nonlinear operators via DeepONet. *Nat. Mach. Intell.* **3**, 218 (2021).
62. Battaglia, P. W. et al. Relational inductive biases, deep learning, and graph networks. *arXiv* 1806.01261 (2018).
63. Bakarji, J., Callahan, J., Brunton, S. L. & Kutz, J. N. Dimensionally consistent learning with Buckingham Pi. *Nat. Comput. Sci.* **2**, 834 (2022).
64. Xie, X. et al. Data-driven discovery of dimensionless numbers and governing laws from scarce measurements. *Nat. Commun.* **13**, 7562 (2022).
65. Brunton, S. L., Proctor, J. L. & Kutz, J. N. Discovering governing equations from data by sparse identification of nonlinear dynamical systems (SINDy). *Proc. Natl Acad. Sci. USA* **113**, 3932 (2016).
66. Pope, S. B. *Turbulent Flows* (Cambridge Univ. Press, 2000).
67. Duraisamy, K., Iaccarino, G. & Xiao, H. Turbulence modeling in the age of data. *Annu. Rev. Fluid Mech.* **51**, 357 (2019).
68. Ling, J., Kurzawski, A. & Templeton, J. Reynolds averaged turbulence modelling using deep neural networks with embedded invariance. *J. Fluid Mech.* **807**, 155 (2016).

69. Akeson, R. L. et al. The NASA Exoplanet Archive: data and tools for exoplanet research. *Publ. Astron. Soc. Pac.* **125**, 989 (2013).
70. Feynman, R. P., Leighton, R. B. & Sands, M. *The Feynman Lectures on Physics* (Addison-Wesley, 1964).
71. Chandrasekhar, S. The maximum mass of ideal white dwarfs. *Astrophys. J.* **74**, 81 (1931).
-

Acknowledgements

We thank the Johns Hopkins Turbulence Database and the NASA Exoplanet Archive for public data access. (*Funding sources and computing acknowledgements to be added.*)

Author contributions

C.H. conceived the method, designed and trained the model, ran the experiments, and wrote the manuscript. J.C. contributed to the benchmark construction, case-study analysis, and manuscript revision. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to C.H. (ch2067@cam.ac.uk). **Supplementary Information** accompanies this paper. **Extended Data** are available with the online version.
